

Exam: T-Test and Group Comparisons

Multiple Choice (Single Select) - 4 Answer Options

1. What is the t-test used for?

- A) Comparing variances
- B) Testing normality
- C) Comparing means of two groups
- D) Testing correlation

2. Which assumption applies to the independent t-test?

- A) Ordinal DV
- B) No outliers
- C) Equal sample size
- D) Normality of DV in both groups

3. What does Levenes test assess?

- A) Independence
- B) Equality of variances
- C) Equality of means
- D) Normality

4. When do you use the paired t-test?

- A) More than two groups
- B) Two independent groups
- C) Two dependent measurements from same subjects
- D) One sample vs. population

5. What is Cohen's d?

- A) A measure of effect size
- B) The t-value
- C) The standard deviation
- D) The correlation

6. When should Welch's test be preferred?

- A) When variances are unequal
- B) When variances are equal
- C) With large samples
- D) With normal distribution

7. What are the degrees of freedom for $n_1=n_2=20$?

- A) 20
- B) 38
- C) 19
- D) 40

8. What does $p < 0.05$ mean?

- A) H_1 is proven
- B) The sample is representative
- C) Reject H_0
- D) The effect is practically relevant

Solutions

1: C

2: D

3: B

4: C

5: A

6: A

7: B

8: C